

Power Electronics Laboratory

Session 5: Three-Phase DC-AC Converter

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Introduction

This laboratory session investigates the operational principles and modulation strategies of a Three-Phase 2-Level Voltage Source Inverter (VSI). The experiment is divided into two main parts:

- **Standard Operation:** Constructing a Sinusoidal PWM (SPWM) circuit to analyze the relationship between switching states, phase voltages, and active power under various modulation indexes.
- **Advanced Modulation Techniques:** Exploring the transition from the linear region to the overmodulation region to observe low-frequency harmonic distortion. Furthermore, the implementation of Third-Harmonic Injection and Min-Max Injection (equivalent to SVPWM) is evaluated. These common-mode injection techniques aim to maximize the DC bus utilization while maintaining a purely sinusoidal load voltage through floating neutral cancellation.

1 Part 1

1.1 Task 1:

State	Sa	Sb	Sc	V _{an}	V _{bn}	V _{ab}	V _{aN}	V _{nN}
0	1	0	0	400	-200	600	600	200
1	1	1	1	200	200	0	600	400
2	0	1	0	-200	400	-600	0	200
3	0	1	1	-400	200	-600	0	400
4	0	0	1	-200	-200	0	0	200
5	1	0	1	200	-400	600	600	400
6	1	1	1	0	0	0	600	600
7	0	0	0	0	0	0	0	0

Table 1: State Table

1.2 Task 2:

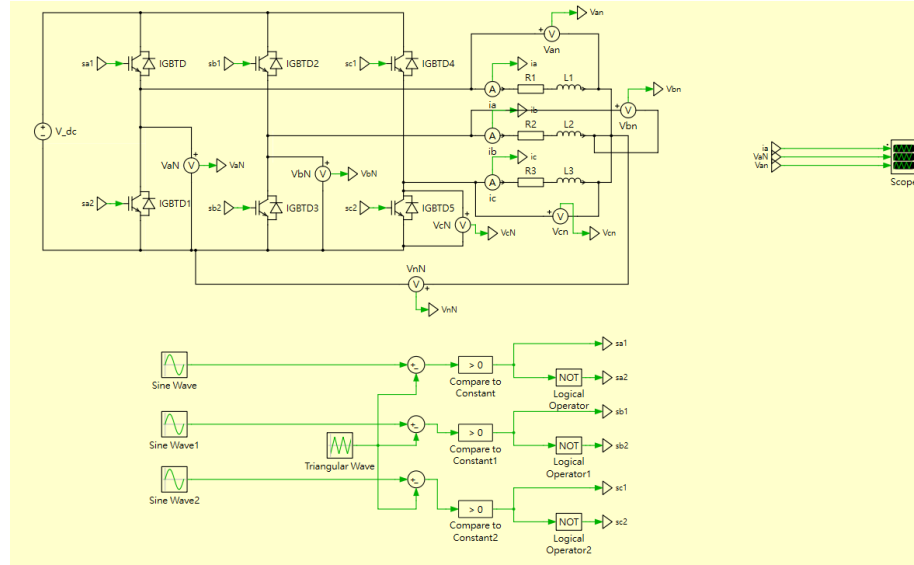


Figure 1: Topology 3-phase 2-level converter

1.3 Task 3:

1.3.1 Calculation of Modulation Indexes (m)

In a 3-phase SPWM voltage source inverter, the line-to-line RMS voltage ($V_{LL,rms}$) relates to the DC bus voltage ($V_{dc} = 600V$) and the modulation index (m)

through the equation:

$$V_{LL,rms} = \sqrt{3} \cdot \frac{m \cdot V_{dc}}{2\sqrt{2}} \approx m \cdot 367.42V$$

By rearranging the formula, the required modulation indexes are calculated as follows:

- For 360V: $m = 360/367.42 \approx \mathbf{0.980}$
- For 300V: $m = 300/367.42 \approx \mathbf{0.816}$
- For 250V: $m = 250/367.42 \approx \mathbf{0.680}$

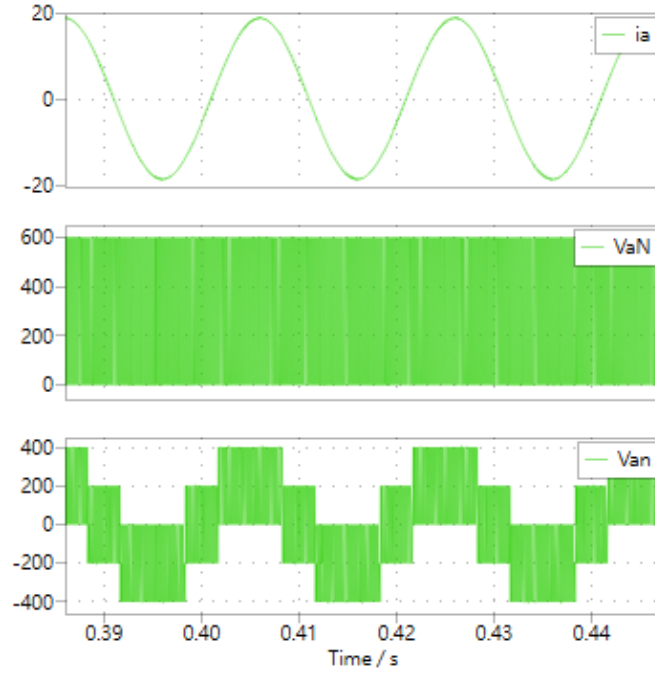


Figure 2: $m = 0.980$

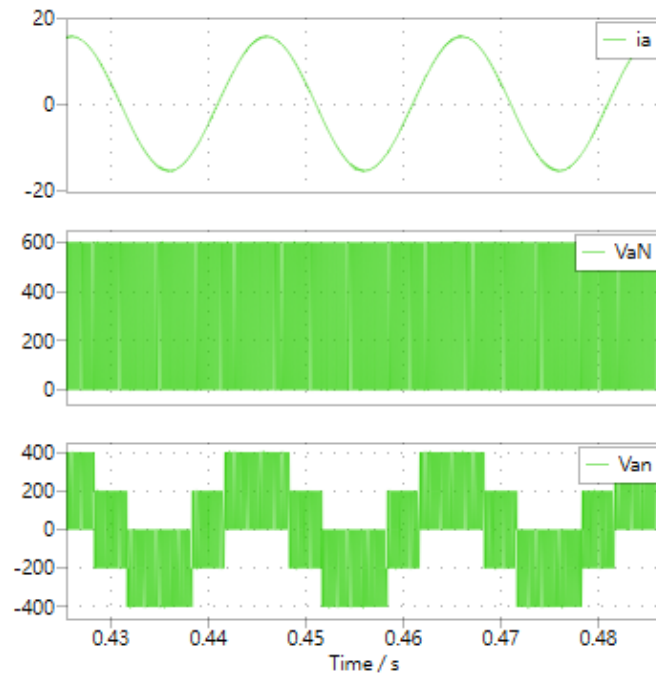


Figure 3: $m = 0.816$

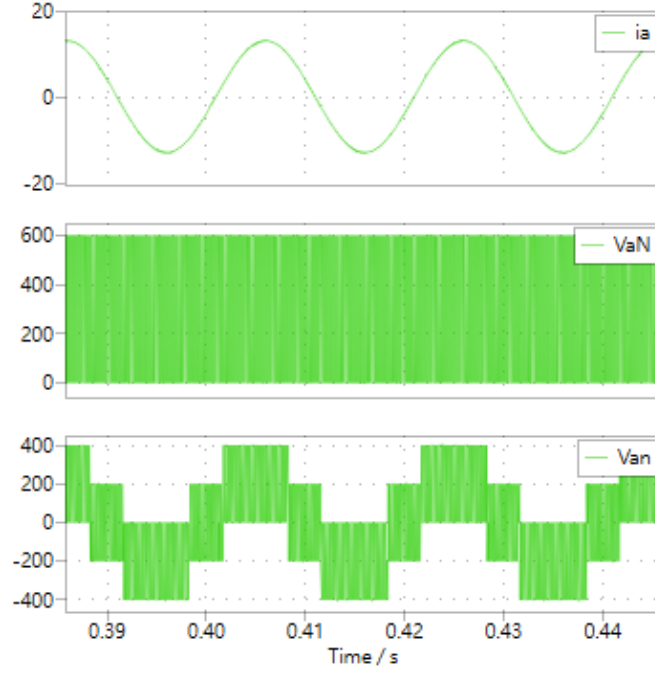


Figure 4: $m = 0.680$

1.3.2 Load Current and Active Power Calculations

Given the fundamental frequency $f = 50\text{Hz}$, a load inductance of $L = 15\text{mH}$, and a resistance of $R = 15\Omega$, the magnitude of the load impedance is $|Z| = \sqrt{15^2 + (2\pi \cdot 50 \cdot 0.015)^2} = \sqrt{15^2 + 4.71^2} \approx \mathbf{15.72\Omega}$. The RMS phase voltage is calculated as $V_{an,rms} = V_{LL,rms}/\sqrt{3}$.

Using these values, the theoretical RMS phase current ($I_{rms} = V_{an,rms}/|Z|$) and the total 3-phase active power ($P = 3 \cdot I_{rms}^2 \cdot R$) are:

- Case 1 ($360\text{V}, m_a = 0.980$): $V_{an,rms} = 207.85\text{V} \rightarrow I_{rms} = \mathbf{13.22\text{A}}, P = \mathbf{7867\text{W}}$

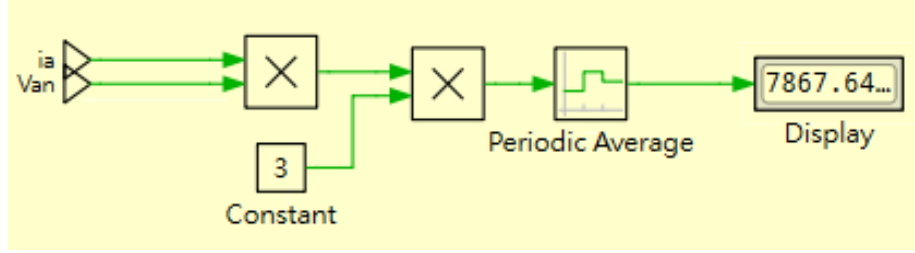


Figure 5: Active Power case 1

- Case 2 (300V, $m_a = 0.816$): $V_{an,rms} = 173.21\text{V} \rightarrow I_{rms} = 11.02\text{A}$, $P = 5454\text{W}$

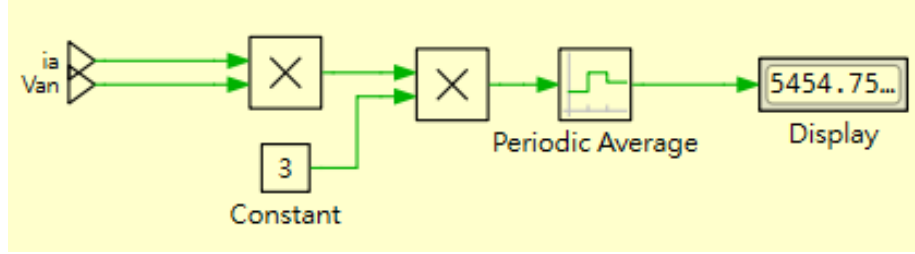


Figure 6: Active Power case 2

- Case 3 (250V, $m_a = 0.680$): $V_{an,rms} = 144.34\text{V} \rightarrow I_{rms} = 9.18\text{A}$, $P = 3788\text{W}$

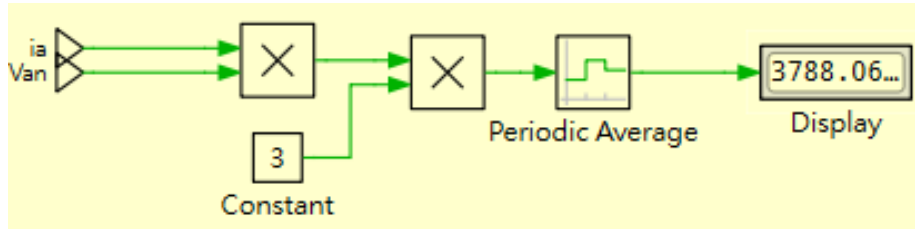


Figure 7: Active Power case 3

1.3.3 Simulation Verification

The converter was simulated in PLECS using the calculated m_a values. The measured output voltage relative to the DC negative bus (v_{aN}) switches strictly between 0V and +600V. In contrast, the load phase voltage (v_{an}) swings symmetrically around 0V with a multi-level stepped waveform. The measured RMS

currents and 3-phase active power in the simulation strongly align with the theoretical calculations, successfully validating the SPWM configuration and the power transfer capability.

1.4 Task 4:

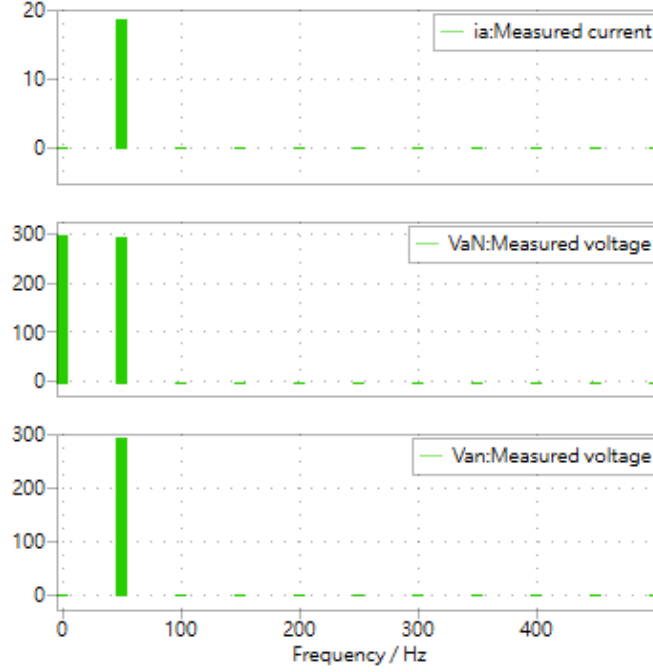


Figure 8: FFT spectrum of $m = 0.98$)

As observed in the FFT spectrum, the voltage relative to the DC negative bus (v_{aN}) contains a massive DC component (0 Hz) equal to exactly 300V, which is $V_{dc}/2$. This occurs because the leg voltage v_{aN} switches strictly between 0V and +600V with a 50% duty cycle at the fundamental zero-crossing, resulting in a constant average offset of 300V.

However, this 300V DC component completely disappears in the load phase voltage (v_{an}) and the line-to-line voltage (v_{ab}). The justification lies in the balanced three-phase star topology:

- Cancellation in Line-to-Line Voltage (v_{ab}): All three converter legs (v_{aN} , v_{bN} , v_{cN}) possess the exact same 300V DC offset. Since the line-to-line voltage is the difference between two legs ($v_{ab} = v_{aN} - v_{bN}$), the common DC offsets perfectly cancel each other out ($300V - 300V = 0V$).

- Cancellation in Phase Voltage (v_{an}): In a balanced star-connected load with a floating neutral, the neutral point voltage (v_{nN}) represents the instantaneous average of the three leg voltages. Consequently, v_{nN} inherently absorbs this 300V common-mode DC offset. The load phase voltage is defined as $v_{an} = v_{aN} - v_{nN}$. Subtracting the neutral point's 300V bias from the leg's 300V bias entirely eliminates the DC component, leaving a pure AC waveform across the load.

2 Part 2

2.1 Task 1:

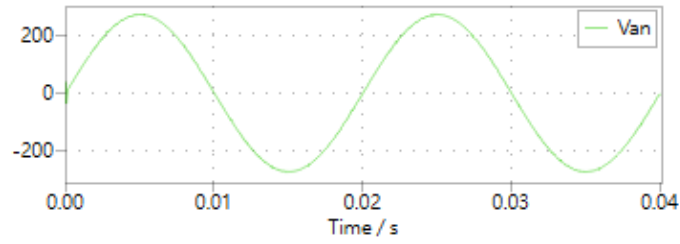


Figure 9: $m = 0.9$

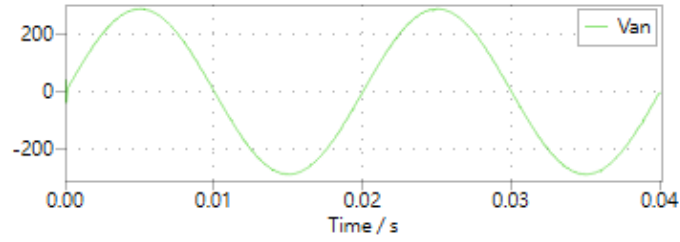


Figure 10: $m = 0.95$

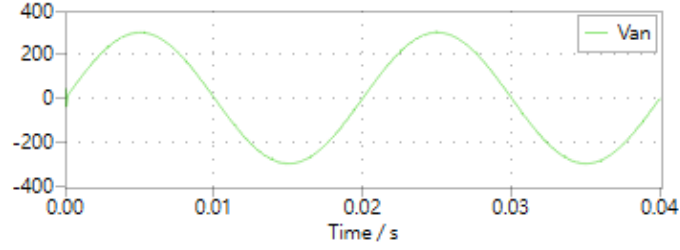


Figure 11: $m = 1.0$

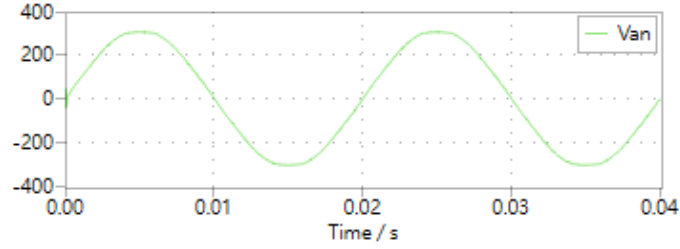


Figure 12: $m = 1.05$

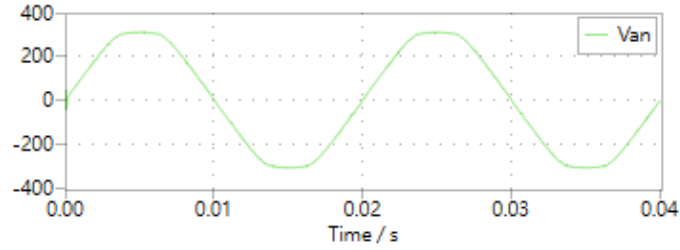


Figure 13: $m = 1.1$

2.1.1 Observation of Low-Frequency Component Waveforms:

By applying a moving average filter tuned to the switching frequency ($f_{sw} = 10\text{kHz}$), the high-frequency switching ripple is removed, revealing the fundamental low-frequency component of the output voltage.

- In the *linear region* ($m = 0.90, 0.95, 1.00$), the filtered output voltage remains a pure, undistorted sinusoidal waveform. The amplitude of the fundamental voltage increases linearly with m .
- In the **over-modulation region** ($m = 1.05, 1.10$), the reference sine wave exceeds the amplitude of the triangular carrier wave. This causes the

PWM modulator to stop switching near the peaks of the reference signal, a phenomenon known as "pulse dropping." As a result, the filtered output voltage waveform becomes clipped (flattened at the top and bottom), gradually transitioning from a pure sine wave toward a square wave.

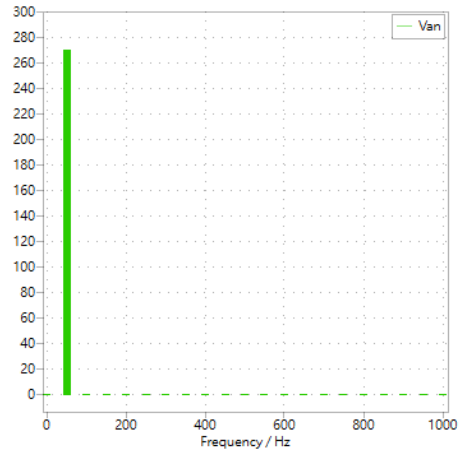


Figure 14: Enter Caption

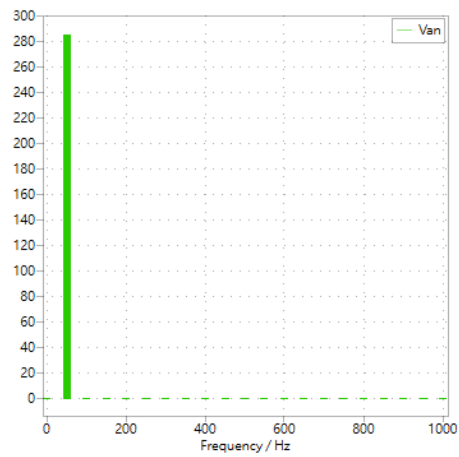


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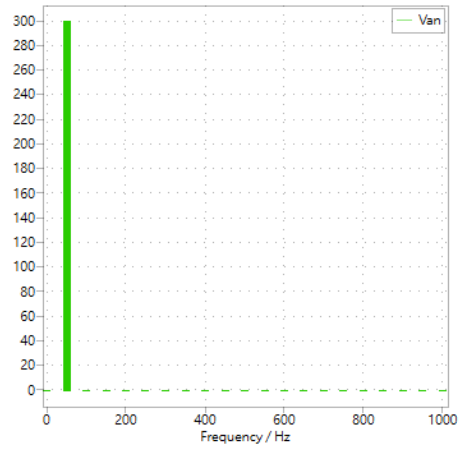


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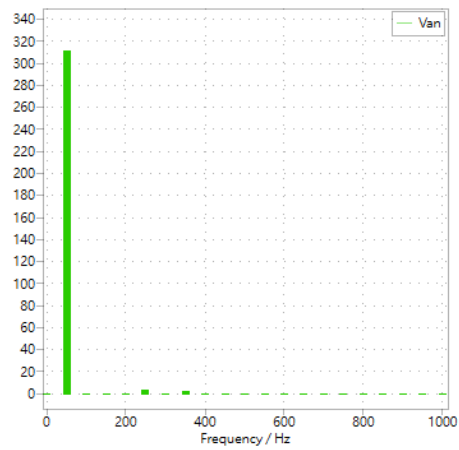


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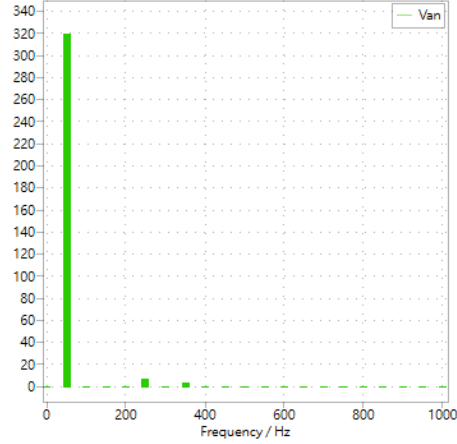


Figure 18: Enter Caption

2.1.2 Comment on Low-Frequency Harmonics (FFT Analysis 0-1kHz):

- For $m \leq 1.0$, the FFT spectrum between 0 and 1 kHz contains almost exclusively the 50Hz fundamental frequency. Low-order harmonics are virtually nonexistent.
- For $m > 1.0$, the clipping effect introduced by overmodulation severely distorts the waveform, which mathematically generates low-frequency harmonics. Due to the symmetrical nature of the waveform and the absence of a neutral return path in the balanced 3-phase load, triplen harmonics (3rd, 9th, 15th) are inherently suppressed. Consequently, the dominant low-frequency harmonics that emerge in the FFT spectrum are the **5th(250 Hz), 7th(350 Hz), and 11th(550 Hz)**. As m_a increases further into overmodulation, the magnitude of these low-order harmonics grows significantly, which deteriorates the power quality and increases harmonic distortion in the load current.

2.2 Task 2:

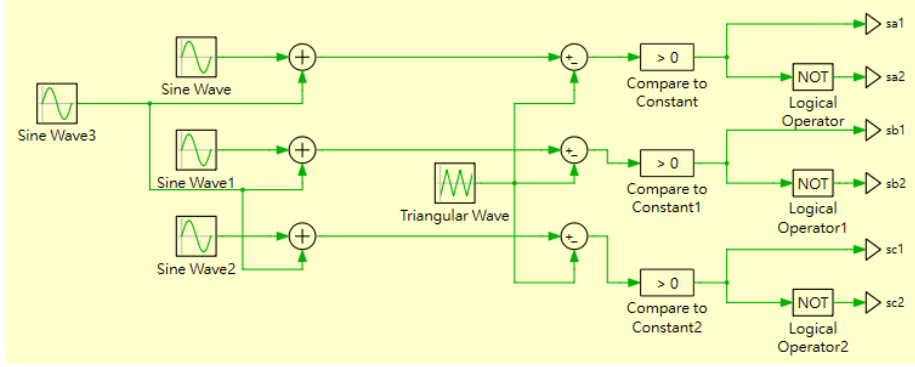


Figure 19: Topology of standard PWM with common-mode component

2.2.1 Theoretical Justification:

By injecting a third-harmonic component into the fundamental reference signal, the peak of the resultant modulating signal is flattened. The injected modulating signal is defined as:

$$m_{new}(t) = M \sin(\omega t) + \frac{M}{6} \sin(3\omega t)$$

- To find the maximum value of this function, we evaluate it at $\omega t = 60^\circ$ (where the peak of the flattened waveform occurs):

$$m_{new,max} = M \sin(60^\circ) + \frac{M}{6} \sin(180^\circ) = M \frac{\sqrt{3}}{2}$$

- To prevent overmodulation and pulse dropping, the peak of the modulating signal must not exceed the triangular carrier amplitude (1.0). Therefore:

$$M \frac{\sqrt{3}}{2} \leq 1 \implies M \leq \frac{2}{\sqrt{3}} \approx 1.154$$

This demonstrates mathematically that the fundamental modulation index (m_a) can be increased to approximately 1.15 without entering the overmodulation region, extending the linear range by 15%.

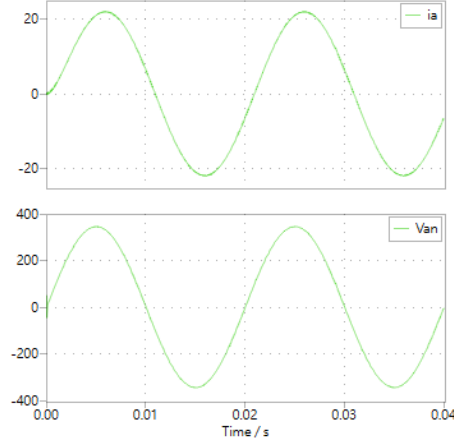


Figure 20: Low-Frequency Output Voltage for Different Modulation Indices

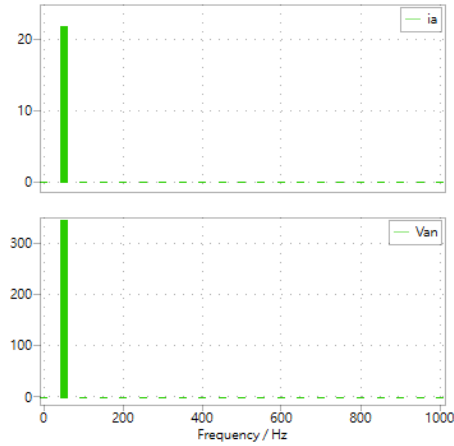


Figure 21: FFT Spectrum of the Output Voltage

2.2.2 Simulation Verification:

The third-harmonic injection was implemented in PLECS by adding a 150Hz common-mode sine wave (amplitude $1.15/6$) to the three-phase 50Hz fundamental references (amplitude 1.15).

- **Modulating Signal:** The resultant reference signal exhibits a saddle shape, effectively keeping its peak within the $[-1, 1]$ bounds.
- **Load Phase Voltage and FFT:** Since the injected third-harmonic is a zero-sequence (common-mode) component, it is identical in all three

phases and is naturally cancelled out in the line-to-line voltages and absorbed by the floating neutral point. The filtered load phase voltage (v_{an}) remains a pure, unclipped sinusoidal waveform. The FFT spectrum verifies that, despite $M = 1.15$, there are no significant low-order harmonics (e.g., 5th or 7th) introduced, successfully avoiding the harmonic distortion typically associated with standard SPWM overmodulation.

2.3 Task 3:

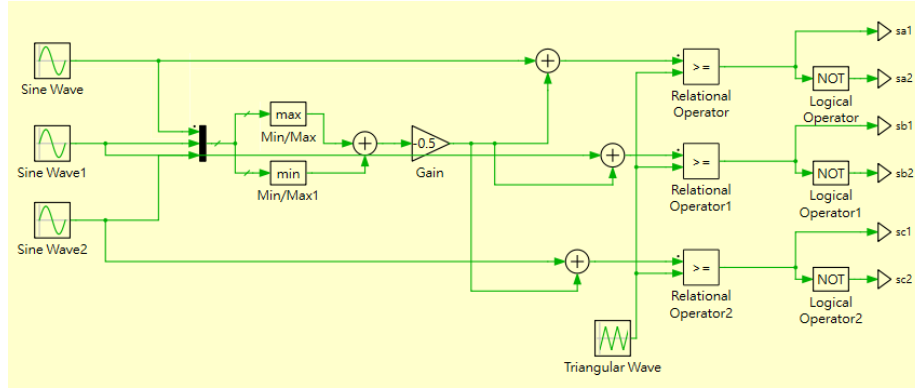


Figure 22: Topology of standard PWM with a new common mode component:

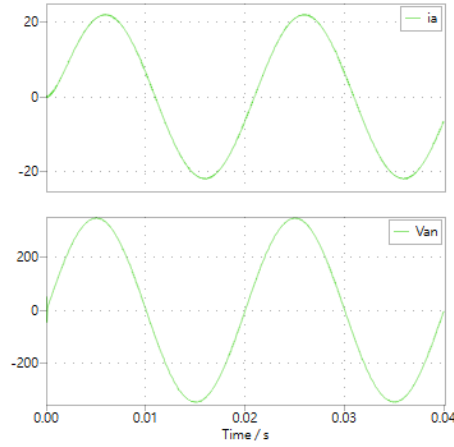


Figure 23: Common-Mode V. (V_n) under Third-Harmonic and Min-Max Injection Techniques

2.3.1 Implementation and Waveform Observation:

The Min-Max injection technique was implemented by calculating the instantaneous common-mode voltage offset using the formula:

$$V_{cm} = -\frac{\max(m_a, m_b, m_c) + \min(m_a, m_b, m_c)}{2}$$

This common-mode signal is then added back to the three fundamental sine waves ($m_{new_x} = m_x + V_{cm}$).

Observing the modulating signal m_{new_a} in the simulation reveals a characteristic "double-hump" (camel-back) waveform. This specific injection mathematically centers the active voltage vectors within the carrier bounds at all times, making it identical to the continuous Space Vector PWM (SVPWM) strategy. As observed in the filtered phase voltage (v_{an}), the output remains perfectly sinusoidal without any low-frequency harmonic distortion, even when the modulation index is pushed beyond 1.0.

Comparison of DC Bus Utilization:

- Standard SPWM: In conventional sinusoidal PWM, the peak of the modulating signal directly corresponds to the fundamental amplitude. The maximum linear modulation index without clipping is $M = 1.0$. The maximum achievable fundamental phase voltage peak is $V_{dc}/2$.
- Min-Max Injection: By flattening the peaks of the modulating waves, the Min-Max technique allows the fundamental modulation index to be increased up to $M = \frac{2}{\sqrt{3}} \approx 1.1547$ before the flattened peaks hit the triangular carrier limits (± 1.0).

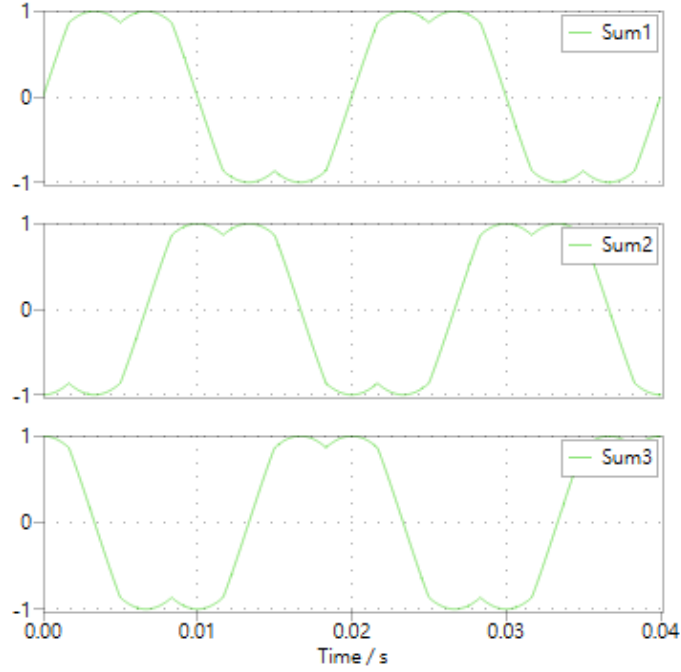


Figure 24: Modulating signal before the Comparator

By observing the modulating signal before the comparator, the characteristic double-hump waveform is clearly visible. The Min-Max injection perfectly bounds the peak of the $M = 1.15$ reference signal within the ± 1.0 carrier limits, proving how the linear modulation range is extended.

Comments: The Min-Max injection technique seamlessly extends the linear operation region. Compared to standard SPWM, it maximizes the available DC bus utilization by 15.47%, allowing the converter to extract more AC voltage from the same DC source while preventing low-order harmonic distortion associated with overmodulation.

2.4 Task 4:

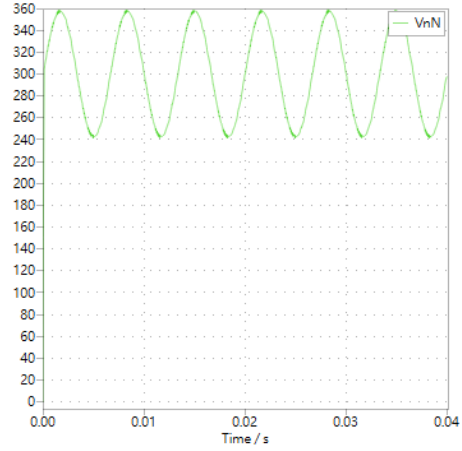


Figure 25: Third-Harmonic injection Common Mode Voltage plot

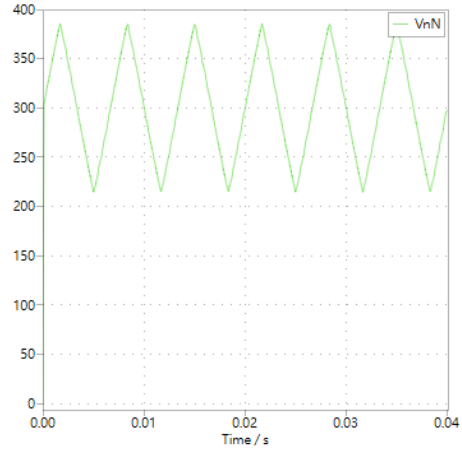


Figure 26: Min-Max injection Common Mode Voltage plot

2.4.1 Observation of Common Mode Voltage (v_{nN}):

When measuring the common-mode voltage v_{nN} (the voltage between the load star point and the negative DC bus) with a moving average filter, it becomes evident where the injected distorted signals reside.

- In the **Third-Harmonic injection**, the low-frequency component of v_{nN} contains the standard 300V DC offset ($V_{dc}/2$) superimposed with a distinct 150Hz sinusoidal ripple.

- In the **Min-Max injection**, v_{nN} displays the 300V DC offset superimposed with a highly distorted, triangular-like ripple that corresponds directly to the injected $-\frac{\max+\min}{2}$ signal.

2.5 Task 5:

2.5.1 Critical Justification (Why load voltage remains perfectly sinusoidal):

Even though heavily distorted signals are injected into the PWM modulating references to maximize DC bus utilization, the load phase voltage (v_{an}) and line currents remain perfectly sinusoidal. This phenomenon is justified by the circuit topology and the nature of the injected signals:

- **Zero-Sequence Nature:** Both the third-harmonic and the Min-Max components are injected identically into all three phases. Therefore, they act strictly as zero-sequence (or common-mode) voltages.
- **Floating Neutral Cancellation:** The load is connected in a balanced star (Y) configuration without a physical neutral wire returning to the source (floating neutral). In such a balanced system, the neutral point voltage (v_{nN}) is mathematically constrained to be the instantaneous average of the three leg voltages (v_{aN}, v_{bN}, v_{cN}). Consequently, the neutral point inherently absorbs the entirety of the injected common-mode signal (along with the $V_{dc}/2$ offset).
- **Differential Output:** The actual voltage applied across the load phase is the differential voltage between the converter leg and the neutral point:

$$v_{an} = v_{aN} - v_{nN}$$

Because the distorted injected signal exists identically in both v_{aN} and v_{nN} , it is perfectly subtracted out in the equation. The common-mode distortion cancels itself completely, leaving only the pure, fundamental 50Hz differential-mode sine wave across the load.

Conclusion

This experiment successfully verifies the characteristics and limitations of different modulation techniques in a 3-phase VSI. The key findings are summarized as follows:

- **SPWM and DC Offset:** In standard SPWM, the $V_{dc}/2$ common-mode DC offset generated by the leg voltages is inherently cancelled out in the line-to-line and load phase voltages due to the balanced star-topology.
- **Over-modulation Consequences:** Operating the inverter in the over-modulation region ($m_a > 1.0$) causes pulse dropping and waveform clipping, which significantly degrades power quality by introducing low-order harmonics (e.g., 5th and 7th).

- **DC Bus Utilization Enhancement:** By flattening the modulating signal peaks, both Third-Harmonic and Min-Max Injection successfully extend the linear modulation range to $M \approx 1.1547$. This yields a 15.47% increase in maximum achievable AC voltage without clipping.
- **Floating Neutral Filtering:** Although advanced modulations inject heavily distorted signals (such as saddle or double-hump waveforms), these signals are strictly zero-sequence. They are entirely absorbed by the floating neutral point, ensuring that the differential output across the load remains a pure fundamental sine wave.